Security Risks and Privacy Concerns Using Generative AI

Foundational Principles of Generative AI

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Overview

Underlying principles of generative models

GANs

VAEs

GAN

Generative adversarial network

Machine learning model

Generate new data that resembles existing data

GAN

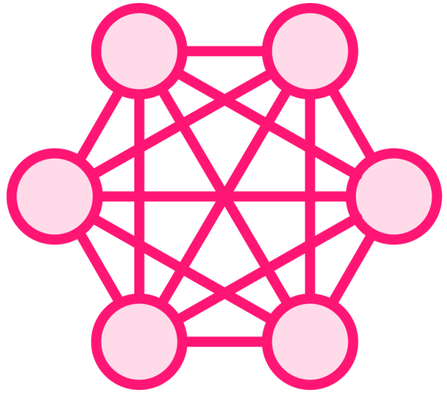
Generator

Discriminator

GAN

Generate realistic data

Difficult to train



Powerful tool

VAE

Variational autoencoders

Probabilistic generative models

Unsupervised learning tasks

Data generation

Realistic and diverse

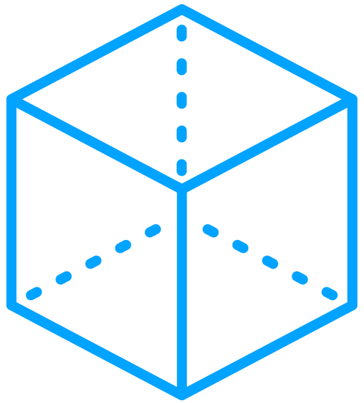
Variety of data

VAE

Encoder

Decoder

VAE



Generate realistic data

Less realistic than GANs

Transparent and Interpretable Generative AI Models

Impact on trust, fairness, and accountability

Trust

Instill trust in users and stakeholders

Clear and understandable explanations for decisions and outputs

Equipped to adhere to ethical guidelines and regulatory requirements

Fairness

Mitigate biases and promote fairness

Biases can be detected and addressed

Promote equitable outcomes

Accountability

Traceability

Identify errors and biases

Transparent and Interpretable Generative AI Models

Enhance user confidence

Help address bias